

185. Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1

185.1 Overview

This clause specifies the 800GBASE-LR1 PMD together with the associated single-mode fiber medium. The optical signal generated by this Physical Medium Dependent (PMD) type is modulated using a dual polarization 16-state quadrature amplitude modulation (DP-16QAM) format suitable for reception by a coherent optical receiver. The PMD provides point-to-point 800 Gb/s Ethernet links over one pair of single-mode fiber, with a reach of up to at least 10 km.

When forming a complete Physical Layer, a PMD shall be connected to the 800GBASE-LR1 Inner FEC as shown in Table 185–1.

Table 185–1—Physical Layer clauses associated with the 800GBASE-LR1 PMD

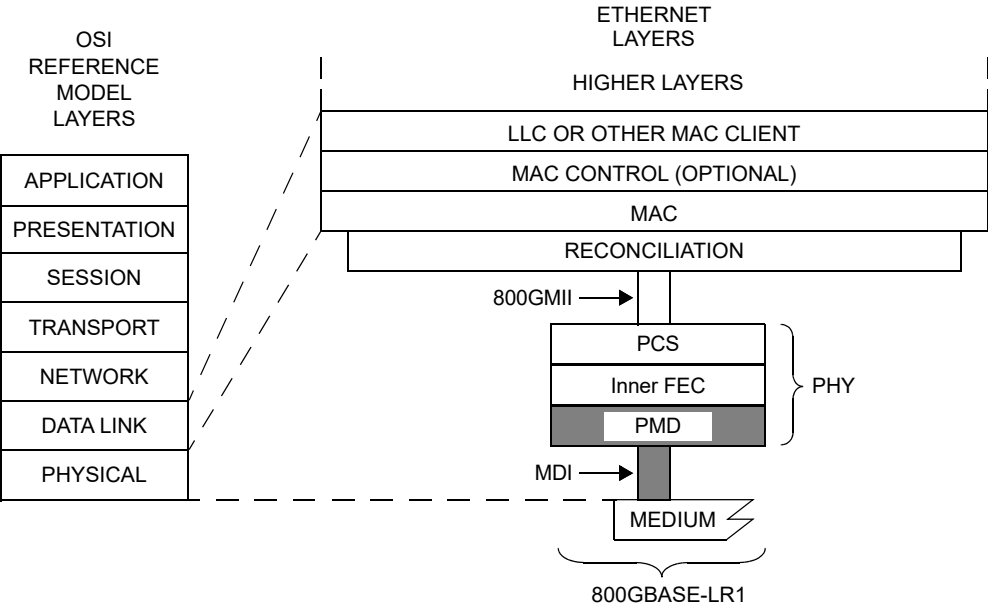
Associated clause	800GBASE-LR1
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Conditional ^c
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a
184—800GBASE-LR1 Inner FEC	Required

^a One or two 800GAUI-n may be instantiated within a 800GBASE-LR1 PHY as described in 176B.7.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If one or two 800GAUI-n are implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.7.1.

Figure 185–1 shows the relationship of the PMD and MDI (shown shaded), along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model. 800 Gigabit Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMD = PHYSICAL MEDIUM DEPENDENT
LR1 = PMD FOR SINGLE MODE FIBER — 10 km

Figure 185–1—800GBASE-LR1 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

185.2 Error ratio allocation

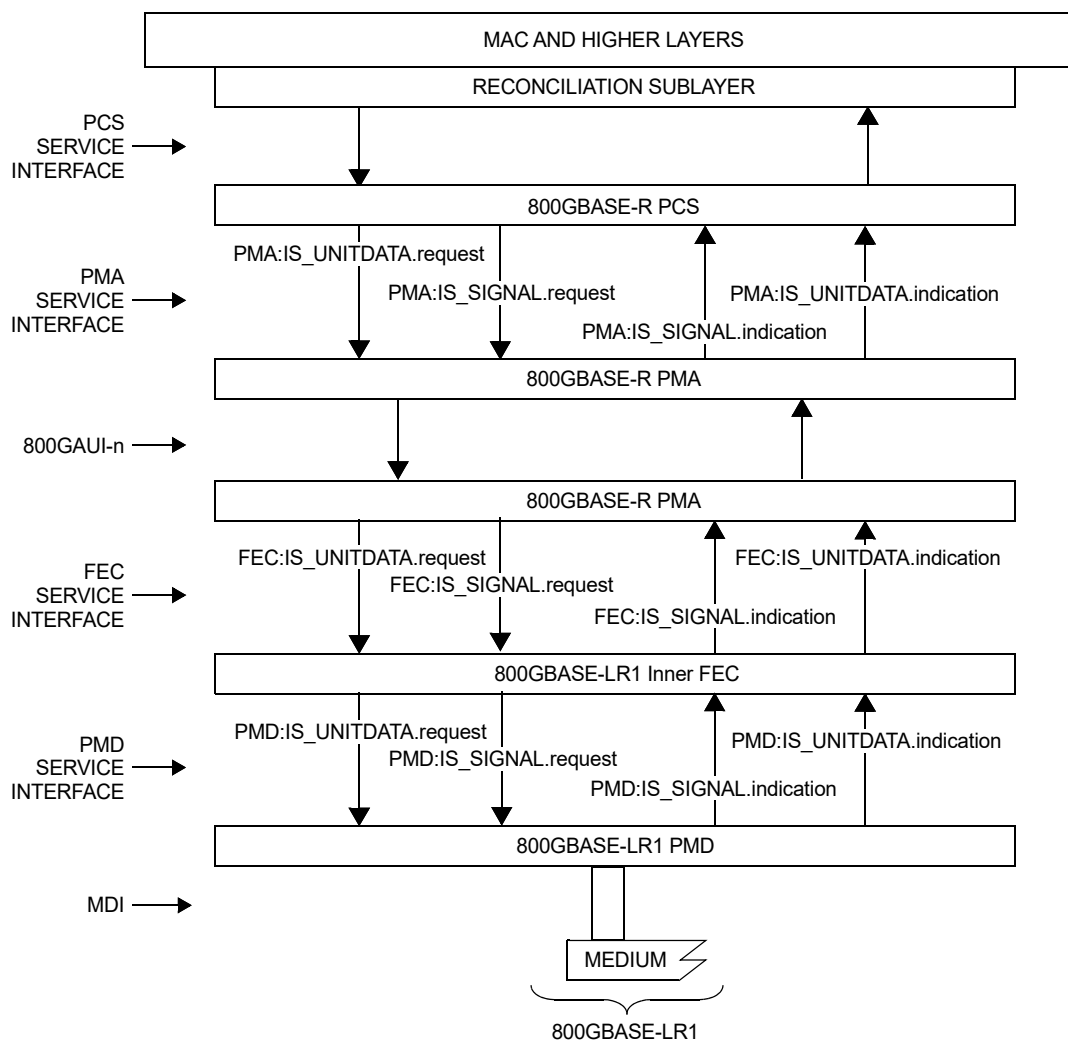
A complete PHY is expected to meet the frame loss ratio specifications in 174A.6.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio (BLER) of 1.45×10^{-11} (see 174A.6), measured at the Inner FEC using the method described in 174A.10, with BER_{added} equal to 6.4×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the BLER of 1.45×10^{-11} (see 174A.6), measured at the PCS using the method described in 174A.11, with BER_{added} equal to 3.2×10^{-5} .

185.3 PMD service interface

This subclause specifies the services provided by the 800GBASE-LR1 PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the Inner FEC sublayer that resides just above the PMD, and the PMD sublayer. The PMD translates the encoded data to and from signals suitable for the specified medium.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 185-2—800GBASE-LR1 inter-sublayer service interfaces including 800GMII Extender

185.3.1 800GBASE-LR1 PMD service interface

The 800GBASE-LR1 PMD service interface uses the following primitives to exchange signals and status information:

PMD:IS_UNITDATA.request
PMD:IS_UNITDATA.indication
PMD:IS_SIGNAL.indication
PMD:IS_SIGNAL.request

185.3.1.1 PMD:IS_UNITDATA.request

The PMD:IS_UNITDATA.request primitive defines the transfer of four analog signals from the 800GBASE-LR1 Inner FEC to the 800GBASE-LR1 PMD (see 184.5.11.1) that are inputs to the DP-16QAM modulator to generate a single optical signal with orthogonal polarizations (see 185.5.2).

185.3.1.1.1 Semantics of the primitive

PMD:IS_UNITDATA.request(tx_signal_xi, tx_signal_xq, tx_signal_yi, tx_signal_yq)

The PMD_UNITDATA.request primitive conveys four analog signals, representing the in-phase (I) and quadrature (Q) components for each of the polarizations (X and Y), via the tx_signal_xi, tx_signal_xq, tx_signal_yi, and tx_signal_yq parameters.

185.3.1.1.2 When generated

The 800GBASE-LR1 Inner FEC generates PMD:IS_UNITDATA.request continuously.

185.3.1.1.3 Effect of receipt

The effect of receipt of the PMD:IS_UNITDATA.request by the 800GBASE-LR1 PMD is defined in 185.5.2.

185.3.1.2 PMD:IS_UNITDATA.indication

The PMD:IS_UNITDATA.indication primitive defines the transfer of four analog signals from the 800GBASE-LR1 PMD to the 800GBASE-LR1 Inner FEC that are the outputs of the coherent DP-16QAM receiver (see 185.5.3.)

185.3.1.2.1 Semantics of the primitive

PMD:IS_UNITDATA.indication(rx_signal_xi, rx_signal_xq, rx_signal_yi, rx_signal_yq)

The PMD:IS_UNITDATA.indication primitive conveys four analog signals, representing the in-phase (I) and quadrature (Q) components for each of the polarizations (X and Y), via the rx_signal_xi, rx_signal_xq, rx_signal_yi, rx_signal_yq parameters, respectively.

185.3.1.2.2 When generated

The 800GBASE-LR1 PMD generates PMD:IS_UNITDATA.indication continuously.

185.3.1.2.3 Effect of receipt

The 800GBASE-LR1 Inner FEC processes the four analog signals as defined in 184.6.

185.3.1.3 PMD:IS_SIGNAL.indication(SIGNAL_OK)

The PMD:IS_SIGNAL.indication primitive is generated by the 800GBASE-LR1 PMD to indicate the status of the receive function via the SIGNAL_OK parameter.

185.3.1.3.1 Semantics of the service primitive

PMD:IS_SIGNAL.indication(SIGNAL_OK)

The SIGNAL_OK parameter corresponds to the variable Global_PMD_signal_detect as defined in 185.5.4. When Global_PMD_signal_detect is 1, SIGNAL_OK shall be assigned the value OK. When Global_PMD_signal_detect is 0, SIGNAL_OK shall be assigned the value FAIL. When SIGNAL_OK is FAIL, the PMD:IS_UNITDATA_i.indication parameter is undefined.

NOTE—SIGNAL_OK = OK does not guarantee that the rx_signal parameters are known to be good. It is possible for a poor quality link to provide sufficient light for a SIGNAL_OK = OK indication and still not meet the frame loss ratio defined in 185.2.

185.3.1.4 PMD:IS_SIGNAL.request(SIGNAL_OK)

The PMD:IS_SIGNAL.request primitive is generated by the 800GBASE-LR1 Inner FEC to indicate the status of the transmit function via the SIGNAL_OK parameter.

185.3.1.4.1 Semantics of the service primitive

PMD:IS_SIGNAL.request(SIGNAL_OK)

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.request provides the status from ISLs above the PMD.

185.3.1.4.2 When generated

The generation by the Inner FEC is defined in 184.4.

185.3.1.4.3 Effect of receipt

The effect of receipt by the PMD is defined in 185.5.11.

185.4 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-LR1 PMD including 2 m of fiber in one direction shall be no more than 16 384 bit times (32 pause_quanta or 20.48 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

185.5 PMD functional specifications

The 800GBASE-LR1 PMD performs the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

185.5.1 Block diagram

A block diagram for the PMD transmit/receive paths is shown in Figure 185–3 and a block diagram of the PMD is shown in Figure 185–4. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 185.8 and 185.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 185.7). Unless specified otherwise, all receiver measurements and tests defined in 185.8 are made at TP3.

While TP2 and the patch cord for transmitter measurements and tests are shown in Figure 185–3, they are not part of the medium. The transmit MDI is connected to the medium for use, or to the patch cord and TP2 for transmitter measurements and tests.

The analog signals use any of the mappings in Table 185–2.

Table 185–2—Allowed analog signals to symbol mappings

Mapping	Tx_X _I	Tx_X _Q	Tx_Y _I	Tx_Y _Q
1	X _I	X _Q	Y _I	Y _Q
2	X _Q	X _I	Y _Q	Y _I
3	X _I	X _Q	Y _Q	Y _I
4	X _Q	X _I	Y _I	Y _Q
5	Y _I	Y _Q	X _I	X _Q
6	Y _Q	Y _I	X _Q	X _I
7	Y _Q	Y _I	X _I	X _Q
8	Y _I	Y _Q	X _Q	X _I

185.5.3 PMD receive function

The PMD receive function shall convert the composite optical signal received from the MDI into four analog streams (Rx_X_I, Rx_X_Q, Rx_Y_I, Rx_Y_Q) passed across the PMD service interface to the Inner FEC via the PMD:IS_UNITDATA.indication primitive (see 185.3.1.2), all according to the receive optical specifications in this clause.

The four analog streams carry a combination of the transmitting Inner FEC Tx_X_I, Tx_X_Q, Tx_Y_I, and Tx_Y_Q signals used by the transmitting PMD to generate the DP-16QAM symbols.

185.5.4 Global PMD signal detect function

The variable Global_PMD_signal_detect is a global indicator of the presence of an optical signal. For the 800GBASE-LR1 PMD, the variable Global_PMD_signal_detect is set to PMD_signal_detect (see 185.5.5).

The state of the Global_PMD_signal_detect variable is conveyed to PMD client sublayers via the PMD service interface (see 185.3).

185.5.5 PMD signal detect function

The PMD signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input. A receiver shall set the variable PMD_signal_detect to either 0 or 1 according to the criteria in Table 185–6.

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of PMD_signal_detect.

Implementations need to provide adequate margin between the input optical power level at which the PMD_signal_detect is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

Table 185–3—Signal detection criteria

Receive conditions	PMD_signal_detect
Average optical power at TP3 ≤ -19 dBm	0
[The input signal is a compliant 800GBASE-LR1 signal] ^a AND [Optical power at TP3 $\geq$ average receive power (min) in Table 185–6]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

185.5.6 PMD reset function

If the variable PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

PMD reset shall reset the RTS function associated with the PMD (see 185.5.11).

185.5.7 Global PMD transmit disable function

The global PMD transmit disable function enables the optical transmitter to be disabled. This function is used by the ready-to-send (RTS) function (see 185.5.11). It may optionally be provided through the control variable Global_PMD_transmit_disable.

If either Global_PMD_transmit_disable is set to 1 or the variable c_tx_mode = QUIET (see 185.5.11), this function shall turn off the optical transmitter so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 185–5.

If a PMD fault is detected (see 185.5.8), then the PMD may set the Global_PMD_transmit_disable variable to one, turning off the optical transmitter.

185.5.8 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault variable to one.

185.5.9 PMD transmit fault function (optional)

If the PMD has detected a local fault on the transmit path, the PMD shall set the PMD_transmit_fault variable to one.

185.5.10 PMD receive fault function (optional)

If the PMD has detected a local fault on the receive path, the PMD shall set the PMD_receive_fault variable to one.

185.5.11 Autonomous path startup (APSU) functions

A PMD shall provide the RTS function using the RTS with squelch state diagram (see 178B.11) as specified in Annex 178B.

The variable c_rts_status is set as defined for FEC:IS_SIGNAL.indication(SIGNAL_OK) in Table 184–1.

When $c_tx_mode = QUIET$, the PMD squelches the output (see 185.5.7).

When c_tx_mode is TRAINING, RTS, or FILL, the PMD transmits PRBS31 encoded by the 800GBASE-LR1 Inner FEC as defined for the test pattern generator in 184.7.1.

The variable c_isl_ready is equal to $c_local_rx_ready$.

185.6 PMD to MDI optical specifications

The operating range for the 800GBASE-LR1 PMD is defined in Table 185–4. A 800GBASE-LR1 compliant PMD operates over single mode fibers meeting the specifications in Table 185–9. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 800GBASE-LR1 PMD that could operate over 12 km would meet the operating range requirement of 2 m to 10 km).

Table 185–4—800GBASE-LR1 operating range

PMD type	Required operating range
800GBASE-LR1	2 m to 10 km

185.6.1 800GBASE-LR1 transmitter optical specifications

The 800GBASE-LR1 transmitter shall meet the specifications defined in Table 185–5 per the definitions in 185.8.

Table 185–5—800GBASE-LR1 transmit characteristics

Description	Reference	Value	Unit
Signaling rate (range)	—	123.6364 ± 50 ppm	GBd
Modulation format	—	DP-16QAM	—
Average launch power (max)	185.8.2	–6	dBm
Average launch power (min) for $ETCC \leq 1$ dB for $1 < ETCC \leq 2.5$ dB	185.8.2	–11.2 –12.2 + ETCC	dBm
Carrier frequency (range)	185.8.3	228.675 ± 20 GHz	THz
Power difference between X and Y polarizations (max)	185.8.4	1.5	dB
Skew between X and Y polarizations (max)	185.8.5	5	ps
Extended transmit constellation closure (ETCC) (max)	185.8.6	2.5	dB
Instantaneous I-Q offset per polarization (max)	185.8.7	–20	dB
Mean I-Q offset per polarization (max)	185.8.8	–26	dB

Table 185–5—800GBASE-LR1 transmit characteristics (*continued*)

Description	Reference	Value	Unit
I-Q amplitude imbalance (mean)	185.8.9	1	dB
I-Q phase error magnitude (max)	185.8.10	5	deg
I-Q quadrature skew (max)	185.8.11	0.75	ps
Transmitter OSNR in a 12.5 GHz resolution bandwidth (min)	185.8.12	40	dB
Average launch power of OFF transmitter (max)	—	–20	dBm
Transmitter reflectance ^a (max)	—	–20	dB
RIN average (max)	—	–145	dB/Hz
RIN peak (max)	—	–140	dB/Hz
Laser linewidth (max)	—	1	MHz
Tx laser frequency slew rate: pre acquisition ^b (max)	—	10	GHz/s
Tx laser frequency slew rate: post acquisition ^b (max)	—	1	GHz/s
Laser relative frequency tracking accuracy	—	±0.9	GHz
Tx clock phase noise: total integrated random jitter, RMS (max)	185.8.13	0.015	UI
Tx clock phase noise: total periodic jitter, peak-to-peak (max)	185.8.14	0.03	UI

^a Transmitter reflectance is defined looking into the transmitter.

^b Acquisition is when the DSP frame is achieved at the LOCK_DONE state of the DSP lock state diagram (Figure 184–9) for both polarizations.

185.6.2 800GBASE-LR1 receive optical specifications

The 800GBASE-LR1 receiver shall meet the specifications defined in Table 185–6 per the definitions in 185.8.

Table 185–6—800GBASE-LR1 receive characteristics

Description	Reference	Value	Unit
Signaling rate (range)	—	123.6364 ± 50 ppm	GBd
Modulation format	—	DP-16QAM	—
Carrier frequency (range)	—	228.675 ± 20 GHz	THz
Damage threshold ^a	—	–2	dBm
Average receive power tolerance (max)	185.8.15	–4	dBm
Average receive power tolerance (min) for $ETCC \leq 1$ dB for $1 < ETCC \leq 2.5$ dB	185.8.15	–17.5 –18.5 + ETCC	dBm
Receiver sensitivity (max) ^b for $ETCC \leq 1$ dB for $1 < ETCC \leq 2.5$ dB	185.8.16	–18 –19 + ETCC	dBm
Receiver reflectance (max)	—	20	dB
Frequency offset between received carrier and local oscillator (max)	—	40	GHz
Polarization dependent loss (max)	—	2	dB
Polarization rotation speed (max)	—	50	krad/s

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level. The receiver does not have to operate correctly at this input power.

^b Receiver sensitivity is optional.

185.6.3 Illustrative link power budget

An illustrative power budget and penalties for 800GBASE-LR1 channels are shown in Table 185–7.

Table 185–7—800GBASE-LR1 illustrative link power budget

Parameter	Value	Unit
Power budget	6.8	dB
Operating distance	10	km
Channel insertion loss ^a	6.3	dB
Maximum discrete reflectance	–27	dB
Allocation for penalties ^b	0.5	dB
Additional insertion loss allowed	0	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 185–4 and cabled optical fiber attenuation of 0.43 dB/km at 1311 nm plus an allocation for connection and splice loss given in 185.7.2.1.

^b Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

185.7 Optical channel characteristics

The fiber optic cabling model is shown in Figure 185–5.

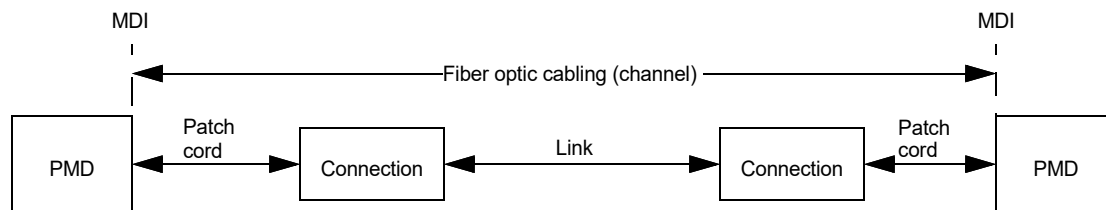


Figure 185–5—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 185–8. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4-2 one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 185–8—Optical channel characteristics

Description	800GBASE-LR1	Unit
Operating distance (max)	10	km
Channel insertion loss ^{a,b} (max)	6.3	dB
Channel insertion loss (min)	0	dB
Positive dispersion (max)	10.1	ps/nm
Negative dispersion (min)	–12.3	ps/nm
Differential group delay (max) ^c	5	ps
Optical return loss (min)	24	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b At 1311 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

185.7.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables containing ITU-T G.652.B (dispersion unshifted), type G.652.D (low water peak, dispersion unshifted), or type G.657.A1, or type G.657.A2 (bend insensitive) fibers, or the requirements in Table 185–9 where they differ.

Table 185–9—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1310	nm
Cabled optical fiber attenuation (max)	0.43 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² × km)

^a The 0.43 dB/km attenuation is provided for Outside Plant cable as defined in ITU-T G.695 Appendix I.

185.7.2 Optical fiber connection

An optical fiber connection, as shown in Figure 185–5, consists of a mated pair of optical connectors.

185.7.2.1 Connection insertion loss

The maximum link distance is based on an allocation of 2 dB total connection and splice loss. For example, this allocation supports four connections with an average insertion loss per connection of 0.5 dB. Connections with different loss characteristics may be used provided the requirements of Table 185–8 are met.

185.7.3 Medium Dependent Interface (MDI)

The 800GBASE-LR1 PMD is coupled to the fiber optic cabling via one fiber per direction at the MDI, being the interface between the PMD and the medium. Examples of an MDI include the following:

- Connectorized fiber pigtail
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 185.5.1, not at the MDI.

185.8 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

185.8.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 185–11 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 185–11 may be used to perform that test. The test patterns used in this clause are shown in Table 185–10.

Table 185–10—Test patterns

Pattern	Pattern description	Defined in
5	Scrambled idle test pattern encoded by the 800GBASE-LR1 Inner FEC	172.2.4.11, 184.5
7	Valid 800GBASE-R signal encoded by the 800GBASE-LR1 Inner FEC	172.2.4, 184.5
8	PRBS31 pattern encoded by 800GBASE-LR1 Inner FEC	184.5.3

Table 185–11—Mapping of parameters to test-patterns and related subclauses

Parameter	Pattern	Related subclause
Average channel output power	5 or 7	185.8.2
Carrier frequency (range)	5 or 7	185.8.3
Power difference between X and Y polarizations	5	185.8.4
Skew between X and Y polarizations	5	185.8.5
ETCC	5	185.8.6
Instantaneous I-Q offset per polarization	5	185.8.7
Mean I-Q offset per polarization	5	185.8.8
I-Q amplitude imbalance	5	185.8.9
I-Q phase error magnitude	5	185.8.10
I-Q quadrature skew	5	185.8.11
Transmitter in-band OSNR	5	185.8.12
Tx clock phase noise: total integrated random jitter	5	185.8.13
Tx clock phase noise: total periodic jitter	5	185.8.14
Average receive power tolerance	5, 7 or 8	185.8.15
Receiver sensitivity	5, 7 or 8	185.8.16

185.8.2 Average channel output power

The average channel output power shall be within the limits given in Table 185–5 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using the test pattern defined in Table 185–11, per the test setup in Figure 53–6.

185.8.3 Carrier frequency (range)

The carrier frequency (range) shall be within the range given in Table 185–5 if measured per IEC 61280-1-3. The transmitter is modulated using the test pattern defined in Table 185–11.

185.8.4 Power difference between X and Y polarizations

The power difference between X and Y polarizations, as defined in Recommendation ITU-T G.698.2, shall be within the limit given in Table 185–5.

185.8.5 Skew between X and Y polarizations

The skew between the X and Y polarizations, as defined in Recommendation ITU-T G.698.2, shall be within the limit given in Table 185–5.

185.8.6 Extended transmitter constellation closure (ETCC)

ETCC is the metric used to define the quality of a 800 Gb/s DP-16QAM transmitter. The ETCC is computed using the test setup and calculation defined in Annex 185A and the parameter values listed in Table 185–12, Table 185–13, and Table 185–14.

The ETCC shall be within the limit given in Table 185–5 if measured using the methods specified.

185.8.6.1 Coherent detector front-end

Parameters for the coherent detector front-end shall meet the values listed in Table 185–12.

Table 185–12—Coherent detector front-end parameters

Description	Value	Unit
3 dB bandwidth (min)	65	GHz
Effective number of bits (<i>ENOB</i>) (min)	5	bits
Oversampling ratio (min)	1.25	—
Local oscillator linewidth (max)	100	kHz
BER_{ref}	1.1×10^{-2}	—

185.8.6.2 Calibrated front-end sampling

The post calibration residuals for the coherent detector front-end shall meet the values listed in Table 185–13.

Table 185–13—Coherent front-end post calibration residuals

Description	Value	Unit
X-Y gain error (max)	0.2	dB
Skew between X-Y polarizations (max)	0.5	ps
I-Q phase error (max)	2	degree
I-Q gain error (max)	0.2	dB
I-Q skew (max)	0.5	ps
Carrier frequency offset (max)	0.1	GHz
Frequency response magnitude variation (max)	1	dB
Frequency response phase variation (max)	0.3	rad
White noise power spectral density variation (max)	0.1	dB
Signal to receiver noise ratio after noise whitening (min)	20	dB

185.8.6.3 Digital signal processing calculation

The digital signal processing parameters for the reference equalizers shall meet the values listed in Table 185–14.

Table 185–14—Digital signal processing parameters

Description	Value
Number of reference equalizer taps	31
Number of reference post-equalizer taps	5

185.8.7 Instantaneous I-Q offset per polarization

The instantaneous I-Q offset per polarization shall be within the limit given in Table 185–5. The instantaneous I-Q offset per polarization is the maximum value per polarization and is calculated as $I\text{-}Q_{\text{offset}} = 10\log_{10}[(I_{\text{mean}}^2 + Q_{\text{mean}}^2)/P_{\text{signal}}]$ with a measurement interval of 1 μs .

185.8.8 Mean I-Q offset per polarization

The mean I-Q offset per polarization shall be within the limit given in Table 185–5. The mean I-Q offset per polarization is the mean value per polarization and is calculated as $I\text{-}Q_{\text{offset}}(\text{mean}) = 10\log_{10}[(I_{\text{mean}}^2 + Q_{\text{mean}}^2)/P_{\text{signal}}]$.

185.8.9 I-Q amplitude imbalance

The I-Q amplitude imbalance is the amplitude difference in dB of the in-phase component I and quadrature component Q of the signal and shall be within the limit given in Table 185–5.

185.8.10 I-Q phase error magnitude

The I-Q phase error magnitude is the phase difference of the in-phase component I and quadrature component Q of the signal, measured relative to local oscillator and shall be within the limit given in Table 185–5.

185.8.11 I-Q quadrature skew

The I-Q quadrature skew is the relative skew between the in-phase component I and quadrature component Q of the signal and shall be within the limit given in Table 185–5.

185.8.12 Transmitter in-band OSNR

The transmitter in-band OSNR shall be within the limit given in Table 185–5. The transmitter in-band OSNR is defined as the ratio of the total signal power in transmitted channel to the highest noise power density (in a 12.5 GHz resolution bandwidth) in the range of the central frequency.

185.8.13 Tx clock phase noise: total integrated random jitter

The Tx clock phase noise: total integrated random jitter, as defined in OIF-800ZR-01.0, Implementation Agreement for 800ZR Coherent Interfaces, section 6.1.213b, shall be within the limits given in Table 185–5.

185.8.14 Tx clock phase noise: total periodic jitter

The Tx clock phase noise: total periodic jitter, as defined in OIF-800ZR-01.0, Implementation Agreement for 800ZR Coherent Interfaces, section 6.1.213c, shall be within the limits given in Table 185–5.

185.8.15 Average receive power tolerance

The average receive power tolerance defines the range of average receiver input power at TP3 over which the BLER requirement in 185.2 is met is met using the test method in either 174A.10 or 174A.11 with parameters provided in Table 185–15. The error mask $H_{\max}(k)$ to be used in the method of 174A.10.4 is provided in Table 185–16. Receiver sensitivity is measured using the patterns listed in Table 185–11.

The conformance test signal meets the requirements for an 800GBASE-LR1 transmitter followed by a channel with the impairments: polarization dependent loss and polarization rotation speed as specified in Table 185–6, and differential group delay and dispersion as specified in Table 185–8.

The ETCC of the transmitter is measured according to 185.9. The ETCC is then used to calculate the minimum average receive power tolerance specified in Table 185–6.

The average receive power tolerance shall meet the limits given in Table 185–6.

Table 185–15—Error ratio parameters

Parameter name	Value
p	1
BLER limit	1.45×10^{-11}
BER_{total}	2.921×10^{-4}
BER_{added}	6.4×10^{-5}
BER_{max}	2.281×10^{-4}

Table 185–16—Receiver error mask

Test symbol errors per test block, k (see 174A.9.5)	Probability $H_{\text{max}}(k)$
1	3.6×10^{-1}
2	2.2×10^{-1}
3	9.2×10^{-2}
4	2.8×10^{-2}
5	7×10^{-3}
6	1.4×10^{-3}
7	2.5×10^{-4}
8	3.9×10^{-5}
9	5.3×10^{-6}
10	6.4×10^{-7}
11	7.1×10^{-8}
12	7.2×10^{-9}
13	6.8×10^{-10}
14	5.9×10^{-11}
15	4.7×10^{-12}
16	3.6×10^{-13}

185.8.16 Receiver sensitivity

Receiver sensitivity is an optional parameter defined as the lowest average receiver input power at TP3 at which the BLER requirement in 185.2 is met using the test method in either 174A.10 or 174A.11 with

parameters provided in Table 185–15. The error mask $H_{\max}(k)$ to be used in the method of 174A.10.4 is provided in Table 185–16. Receiver sensitivity is measured using the patterns listed in Table 185–11.

The conformance test signal meets the requirements for an 800GBASE-LR1 transmitter followed by an optical attenuator.

The ETCC of the transmitter is measured according to 185.9. The ETCC is then used to calculate the maximum receiver sensitivity specified in Table 185–6.

185.9 Safety, installation, environment, and labeling

185.9.1 General safety

All equipment subject to this clause shall conform to the general safety requirements in J.2.

185.9.2 Laser safety

800GBASE-LR1 optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product’s laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.¹⁸

185.9.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

185.9.4 Environment

Normative specifications in this clause shall be met by a system integrating a 800GBASE-LR1 PMD over the life of the product while the product operates within the manufacturer’s range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause are met.

¹⁸A host system that fails to meet the manufacturer’s requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

185.9.5 Electromagnetic emission

A system integrating a 800GBASE-LR1 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

185.9.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

185.9.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-LR1).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 185.9.2. Protocol implementation conformance statement (PICS) proforma for Clause 185, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1.¹⁹

185.10 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 185–17 and Table 185–18. Additional variables associated with the APSU functions are listed in Table 178B–5 and Table 178B–6.

If the MDIO interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 185–17 and Table 185–18.

If the MDIO interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 185–17—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD reset	185.5.6	1.0.15	45.2.1.1.1
Global PMD transmit disable	185.5.7	1.9.0	45.2.1.8.7

¹⁹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

Table 185–18—PMD status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD fault	185.5.8	1.1.7	45.2.1.2.3
PMD transmit fault	185.5.9	1.8.11	45.2.1.7.4
PMD receive fault	185.5.10	1.8.10	45.2.1.7.5
Global PMD signal detect	185.5.4	1.10.0	45.2.1.9.7
PMD signal detect	185.5.5	1.10.1	45.2.1.9.6

185.11 Protocol implementation conformance statement (PICS) proforma for Clause 185, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1²⁰

185.11.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 185, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

185.11.2 Identification

185.11.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

185.11.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 185, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
-------------------	--

²⁰Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

185.11.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
LR1	800GBASE-LR1	185.6	Device supports requirements for 800GBASE-LR1 PHY	O.1	Yes [] No []
TP1	Reference point TP1 exposed and available for testing	185.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
TP4	Reference point TP4 exposed and available for testing	185.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
DC	Delay constraints	185.4	Device conforms to delay constraints	M	Yes []
MD	MDIO capability	185.10	Registers and interface supported	O	Yes [] No []

185.11.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1

185.11.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 800GBASE-R PCS and PMA, and 800GBASE-LR1 Inner FEC	185.1		M	Yes []
F2	Integration with management functions	185.1		O	Yes [] No []
F3	Error ratio allocation	185.2	Meets error ratio allocation specified in 185.2	M	Yes []
F4	Transmit function	185.5.2	Conveys analog data from PMD service interface to MDI	M	Yes []
F5	Receive function	185.5.3	Conveys analog data from PMD service interface to MDI	M	Yes []
F6	Global signal detect	185.5.4	Report to the PMD service interface the message PMD:IS_SIGNAL.indication (SIGNAL_OK)	O	Yes [] No []
F7	PMD reset function	185.5.5	Resets the PMD sublayer	O	Yes [] No []

Item	Feature	Subclause	Value/Comment	Status	Support
F8	Global PMD transmit disable function	185.5.7	Disables the optical transmitter with the Global_PMD_transmit_disable variable	MD:O	Yes [] No [] N/A []
F9	PMD fault function	185.5.8	Sets PMD_fault to one if a local fault is detected	MD:O	Yes [] No [] N/A []
F10	PMD transmit fault function	185.5.9	Sets PMD_transmit_fault to one if a local fault is detected	MD:O	Yes [] No [] N/A []
F11	PMD receive fault function	185.5.10	Sets PMD_receive_fault to one if a local fault is detected	MD:O	Yes [] No [] N/A []

185.11.4.2 PMD to MDI optical specifications for 800GBASE-LR1

Item	Feature	Subclause	Value/Comment	Status	Support
LR1	Transmitter meets specifications in Table 185–6	185.6.1	Per definitions in 185.8	M	Yes []
LR2	Receiver meets specifications in Table 185–7	185.6.2	Per definitions in 185.8	M	Yes []

185.11.4.3 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	185.8	2 m to 5 m in length	M	Yes []
OM2	Average channel output power	185.8.2	Per IEC 61280-1-1	M	Yes []
OM3	Carrier frequency (range)	185.8.3		M	Yes []
OM4	Power difference between X and Y polarizations	185.8.4		M	Yes []
OM5	Skew between X and Y polarizations	185.8.5		M	Yes []
OM6	ETCC	185.8.6		M	Yes []
OM7	Instantaneous I-Q offset per polarization	185.8.7		M	Yes []
OM8	Mean I-Q offset per polarization	185.8.8		M	Yes []
OM9	I-Q amplitude imbalance	185.8.9		M	Yes []
OM10	I-Q phase error magnitude	185.8.10		M	Yes []
OM11	I-Q quadrature skew	185.8.11		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
OM12	Transmitter in-band OSNR	185.8.12		M	Yes []
OM13	Tx clock phase noise: total integrated random jitter	185.8.13		M	Yes []
OM14	Tx clock phase noise: total periodic jitter	185.8.14		M	Yes []
OM15	Average receive power tolerance	185.8.15		M	Yes []

185.11.4.4 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	185.9.1	Conforms to J.2	M	Yes []
ES2	Laser safety	185.9.2	Conforms to Hazard Level 1 laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes []
ES3	Temperature, humidity, and handling	185.9.6	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes []

185.11.4.5 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	185.7.1	Meets the requirements in Table 185–9	M	Yes []
OC2	MDI requirements	185.7.3	Meets IEC 61753-1 and IEC 61753-021-02	INS:M	Yes [] N/A []

185.11.4.6 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	185.10		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	185.10		MD:M	Yes [] N/A []